

Yat (Ѣ) as a Marker of PIE Vowel Lability: Evidence from a 74-Word Corpus and Early Slavic Loanwords

Marko Jevremović

Independent researcher, Belgrade, Serbia

marko@asemic.me

Abstract

The Old Church Slavonic letter yat (Ѣ, ѣ) represents Proto-Slavic **ě*, a vowel unique among Slavic phonemes for producing four distinct modern reflexes: Ekavian /e/, Ikavian /i/, Ijekavian /ije/, and Bulgarian /ja/. We demonstrate that this four-way divergence is not a Slavic-internal accident but a continuation of a deeper pattern: yat systematically marks Proto-Indo-European (PIE) phonological positions where the vowel nucleus was complex—either a laryngeal sequence (**eh₁*) or a diphthong (**ey*, **oy*, **ay*). We construct a 74-word corpus of yat-bearing OCS words with established PIE etymologies and quantify cross-IE vowel divergence at each yat position using a formal Lability Score (LS). Of 67 testable entries, 91% show at least moderate cross-IE vowel lability (mean LS = 3.4), compared with 2.64 for a control set of 53 non-yat **e* words ($p < 0.001$). An independent loanword test examining 45 early borrowings into Proto-Slavic confirms that **ai* diphthongs and long **ē* in donor languages were mapped to yat with 98% accuracy, while other vowel types were never assigned yat (Fisher’s exact test, $p < 0.0001$). A test for correlation between PIE source group and dialect-internal yat reflex exceptions yields a negative result, indicating that the two PIE sources had fully merged before intra-Slavic divergence began. We argue that yat functions as a graphemic marker of inherited PIE vowel instability—a “written laryngeal” that captures the phonological class where IE branches systematically disagree on vowel quality.

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1 Introduction

The letter yat (Ѣ, ѣ) occupies a singular position in Slavic historical phonology. Created as part of the Glagolitic script by Saints Cyril and Methodius in the 9th century CE and subsequently transferred to the Cyrillic alphabet, yat represented the Proto-Slavic vowel **ě*, whose precise phonetic value—likely a mid front vowel [æ] or a long open-mid front vowel—has been debated for over a century. What makes yat unique is not its sound but its behavior: it is the *only* Proto-Slavic vowel to have produced *four* fundamentally different reflexes across the modern Slavic dialect continuum:

- Ekavian /e/ (standard Serbian, Czech, Slovak, etc.)
- Ikavian /i/ (parts of Croatia, Bosnia, some Čakavian dialects)
- Ijekavian /ije/ or /je/ (standard Bosnian, Croatian, Montenegrin)
- Bulgarian /ja/ (or /e/ after palatals)

No other Proto-Slavic vowel exhibits this degree of internal divergence. The vowels **a*, **o*, **u*, **i*, and plain **e* are all reflected with essential uniformity across Slavic; only **ě* fragments into four competing outcomes.

The standard account of yat’s PIE origins is well established. Proto-Slavic **ě* arose from two main PIE sources: (1) a laryngeal sequence **eh₁*, where the first laryngeal lengthened and colored the preceding vowel, yielding a long **ē* that eventually became yat; and (2) PIE diphthongs **ey*, **oy*, and **ay*, which monophthongized to **ě* in Slavic (see Shevelov 1964, Lunt 2001, Derksen 2008). Both sources are attested in well-established etymological correspondences.

What has not been systematically examined, however, is the *cross-IE* behavior of these same roots. We propose that the connection between yat’s PIE source and its modern divergence is not coincidental: the two PIE sources of yat—laryngeal sequences and diphthongs—are precisely the phonological configurations that produce *systematically different vowel outcomes* across the Indo-European branches. We call this property *vowel lability*: the tendency for a given PIE position to be reflected as different vowel qualities in different daughter languages. Our central claim is:

Yat (Ѣ) marks PIE phonological positions where the vowel nucleus was a “complex nucleus” (laryngeal sequence or diphthong), producing inherent cross-IE vowel lability. The four Slavic yat reflexes are a continuation of the same divergence that separates Latin from Greek from Sanskrit from Germanic at these positions.

This paper presents three lines of evidence for this claim. First, we construct a 74-word corpus of yat-bearing OCS words with established PIE etymologies and show that 91% exhibit significant cross-IE vowel lability (§4). Second, we test the hypothesis independently using 45 early loanwords into Proto-Slavic from Germanic, Latin, Iranian, and other sources, demonstrating that yat was assigned to borrowed words if and only if the source vowel belonged to the same phonological class (§5). Third, we test whether the two PIE source groups (laryngeal vs. diphthong) produce different patterns of dialect-internal exceptions, finding that they do not—a negative result that nonetheless clarifies the chronological boundary between cross-IE and intra-Slavic divergence (§6).

2 Background

2.1 The Two PIE Sources of Yat

The Proto-Slavic vowel **ě* (yat) derives from two distinct PIE configurations, both well supported by comparative evidence:

Source A: Laryngeal sequence **eh₁*. When PIE **e* preceded the first laryngeal **h₁*, the sequence **eh₁* produced a long vowel **ē* in most daughter languages. In Slavic, this long **ē* developed into **ě*. Examples include **d^heh₁-* ‘to put, do’ (> дѣло ‘deed’), **seh₁-men-* ‘to sow’ (> сѣмя ‘seed’), and **meh₁-* ‘to measure’ (> мѣра ‘measure’). The laryngeal is not preserved as a segment in any attested IE language but leaves characteristic traces: vowel lengthening, specific ablaut patterns, and the particular quality of the resulting long vowel. Different daughter languages resolve the laryngeal trace differently: Latin typically preserves long *ē*, Germanic develops *æ* (later *ā* in some branches), and Indo-Iranian merges **ē* with **ā*.

Source B: Diphthongs **ey*, **oy*, **ay*. PIE falling diphthongs containing a front glide underwent monophthongization in Proto-Slavic, with the result merging with the laryngeal-derived **ē* as **ě*. Examples include **weyd-* ‘to see, know’ (> вѣдѣти ‘to know’), **sneyg^{wh}-* ‘to snow’ (> снѣгъ ‘snow’), and **k^woyneh₂* ‘payment’ (> цѣна ‘price’). Crucially, other IE branches monophthongize these diphthongs in *radically different ways*: Greek may preserve *oi* or *ei*, Latin develops *ī* or *oe*, Gothic preserves *ai*, Old English produces *ā*, and so on.

2.2 The “Complex Nucleus” Concept

Both PIE sources of yat share a structural property: neither is a simple short vowel. A laryngeal sequence **eH* is a vowel-plus-consonant complex that resolves as a long or colored vowel; a diphthong **ey/*oy* is a vowel-plus-glide sequence. We use the term **complex nucleus** to refer to any PIE syllable nucleus that contains more phonological material than a simple short monophthong.

The key insight is that complex nuclei offer daughter languages *more material to resolve differently*. A simple PIE **e* leaves little room for variation: it surfaces as *e* in Latin, Greek, Slavic, and most of Germanic (with regular ablaut producing *o* in some environments). But a complex nucleus like **ey* provides multiple resolution pathways—the *e* and *y* components can merge in different directions, yielding *ī*, *ei*, *ai*, *e*, or *i* depending on the branch. Similarly, **eh₁* can yield long *ē*, fronted *æ*, backed *ā*, raised *ī*, or reduced *e* depending on the phonological history of each branch.

This is why yat-class words show cross-IE vowel lability while non-yat words do not. The lability is inherent in the PIE source, not in the Slavic development.

2.3 Previous Work

The PIE sources of yat are well established in the standard handbooks (Shevelov 1964, Lunt 2001, Derksen 2008, Fortson 2010). The internal Slavic development of **ě* and its dialect reflexes have been extensively studied (Ivić 1958, Belić 1905, Jakubinskij 1925, Meyer 1926). However, to our knowledge, no previous work has systematically

quantified the *cross-IE* vowel divergence at yat positions or connected it explicitly to the PIE complex-nucleus source.

3 Methodology

3.1 Corpus Construction

We compiled a corpus of 74 Old Church Slavonic words containing yat (ѣ), drawn from three primary sources: Derksen’s *Etymological Dictionary of the Slavic Inherited Lexicon* (2008), Wiktionary Proto-Slavic and PIE reconstruction pages, and the Church Slavonic textbook by Ranković (2015). Selection criteria required that each word (a) contain yat in OCS orthography and (b) have at least one proposed PIE etymology in a published or standard reference source. Four words with no established PIE etymology (тѣло ‘body’, гнѣвъ ‘wrath’, желѣзо ‘iron’, пѣськъ ‘sand’) were included in the corpus but classified as untestable.

The final corpus comprises 69 inherited vocabulary items and 5 loanwords (хлѣбъ ‘bread’, цѣсарь ‘emperor’, хлѣвъ ‘cowshed’, лѣкъ ‘medicine’, and стрѣла ‘arrow’, the last with debated borrowing direction).

3.2 The Lability Score

To quantify cross-IE vowel divergence, we define:

Lability Score (LS) = the number of distinct vowel qualities attested at the yat-equivalent position across IE branches, counting each branch once.

For each corpus entry, we identified the specific vowel at the position corresponding to Slavic ѣ in cognate forms across Latin, Greek, Sanskrit, Germanic (Gothic, OHG, OE), Baltic (Lithuanian, Latvian), Celtic (Old Irish, Welsh), and Iranian (Avestan). Each branch contributes at most one vowel quality. We classified each entry as:

- **MAX** (LS 5–7): four or more distinct vowels across branches
- **STR** (LS 3–4): three or more with clear divergence
- **MOD** (LS 2): two to three distinct variants
- **LIM** (LS 1): very limited cross-IE data
- **N/A** (LS 0): untestable (no PIE etymology or no surviving cognates)

3.3 Divergence Types

Not all cross-IE vowel divergence is equal. A word with LS = 4 because four branches independently resolved a single diphthong to different monophthongs is qualitatively different from LS = 4 because four branches lexicalized four different ablaut grades. To address this, we distinguish four divergence types:

Type N (Nucleus resolution): Different branches resolve the *same* complex nucleus (**eh₁*, **ey*, **oy*, **ay*) from one ablaut grade to different simple vowels. The divergence is inherent in the ambiguity of the nucleus. *This is the mechanism predicted by the hypothesis for yat words.*

Type A (Ablaut-grade divergence): Different branches lexicalized different PIE ablaut grades (e/o/zero/lengthened) of the same root. The vowels differ because the inputs differ, not because one input was ambiguous.

Type R (Regular merger): A branch-wide sound law shifts the vowel predictably. The most common instance is the Indo-Iranian **e* → *a* merger, which affects *all* PIE **e* words in Sanskrit/Avestan and is therefore not position-specific.

Type C (Consonant-conditioned): A nearby consonant—especially a labiovelar (**g^w*, **k^w*)—colors the vowel differently in different branches. The root vowel is simple **e*, but consonant interactions create secondary divergence.

Each corpus entry can be classified by its **dominant divergence type**—the mechanism responsible for the largest share of its cross-branch vowel differences. We apply this classification to both the yat corpus and the control group in §4.6.

3.4 Classification by PIE Source

Each corpus entry was classified into one of six groups based on the PIE source of its yat:

- **Group A** — Laryngeal **eh₁* (18 entries)
- **Group B** — Diphthong **ey*/**oy*/**ay* (26 entries)
- **Group C** — Mixed, debated, or secondary (13 entries)
- **Group S** — Yat in stative suffix **-eh₁-*, not in the root (5 entries)
- **Loanwords** (5 entries)
- **Unknown** — No PIE etymology established (4 entries)

Group S is noteworthy: five verbs (грѣти ‘to heat’, дрѣмати ‘to doze’, имѣти ‘to have’, болѣти ‘to be sick’, терпѣти ‘to endure’) bear yat not in the root but in the stative verbal suffix **-eh₁-*. This suffix is itself a laryngeal-derived complex nucleus, so the prediction still applies—but the yat is grammatical rather than lexical.

3.5 Control Group

As a control, we assembled a set of 53 PIE simple **e* words—those where the root vowel is a plain short **e* without any laryngeal or diphthongal component—and applied the same Labiality Score methodology. Words were excluded if the OCS form contains yat (e.g., мрѣти, дрѣво, берегъ), if the PIE root contains a front diphthong or post-vocalic laryngeal at the measured position (e.g., *vesъ* from **weyk^j-*), or if the Slavic form reflects an o-grade or u-grade rather than the e-grade.

The control group spans verbs (берати, плести, пещи, тесати, жечъ, etc.), kinship terms (жена, деверъ, зять), numerals (десять, шесть, седмь), body/nature words (небо, земля, медъ, перо), and adjectives (ветъхъ, теплъ, зеленъ). The full list of 53 entries with cognate data is provided in the supplementary materials.

Table 1 compares the two distributions.

Table 1: Lability Score comparison: yat corpus vs. non-yat **e* control group.

Metric	Yat corpus	Control (<i>*e</i>)
<i>n</i> (testable)	67	53
Mean LS	3.4	2.64
Median LS	3	3
% with $LS \geq 4$	52%	15%
% with $LS = 1$	0%	6%
Modal LS	3–4	2

Critically, the difference is not merely quantitative. Control words that achieve high LS scores (≥ 4) do so through qualitatively different mechanisms—ablaut alternation across lexicalized grades (e.g., земля, $LS = 4$), labiovelar consonant coloring (e.g., жена, $LS = 4$), or non-yat diphthongs like **ew* (e.g., новъ, $LS = 5$ from **néwos*). In yat words, by contrast, a *single* ablaut grade with a complex nucleus produces divergent outcomes because the nucleus itself is inherently ambiguous. The modal control LS of 2 reflects the baseline Indo-Iranian **e* $\rightarrow$ *a* merger, which affects all PIE **e* words uniformly and is not position-specific.

4 Results

4.1 Corpus Overview

Of the 74 corpus entries, 67 are testable (i.e., have established PIE etymologies with surviving cognates in at least two non-Slavic branches). Of these 67, **61 (91%) show at least moderate cross-IE vowel lability** ($LS \geq 2$). This rate is virtually identical to the 92% observed in an earlier 29-word pilot corpus, confirming that the pattern is robust and does not dilute as the corpus expands.

4.2 Representative Entries

Table 2 presents 20 representative entries spanning all source groups, with compact lability data.

4.3 Lability Statistics

Table 3 summarizes the lability distribution across the full corpus.

The mean Lability Score for testable yat entries is **3.4**, compared with **2.64** for the 53-word non-yat **e* control group (see Table 1). A Welch *t*-test on the full distributions yields $p < 0.001$, confirming that the difference is statistically significant. The distributions also differ qualitatively: 52% of yat words achieve $LS \geq 4$, versus only 15% of control words, while no yat word has $LS = 1$ (compared with 6% of controls).

Table 2: Representative yat-bearing words with PIE source and labiality classification.

#	OCS word	Meaning	PIE root	Grp	LS	Conf
1	дѣло	deed	*d ^h eh ₁ -	A	MOD	H
2	сѣмя	seed	*seh ₁ -men-	A	STR	H
3	мѣра	measure	*meh ₁ -	A	STR	H
5	вѣтръ	wind	*h ₂ weh ₁ -nt-	A	MAX	H
8	вѣдѣти	to know	*weyd-	B	MAX	H
9	снѣгъ	snow	*sneyg ^{wh} -	B	MAX	H
10	цвѣтъ	flower	*k ^h weyt-	B	STR	H
12	цѣна	price	*k ^w oynéh ₂	B	MAX	H
14	гнѣздо	nest	*ni-sd-ó-	B	STR	H
17	млѣко	milk	*h ₂ melǵ-	C	MAX	M
22	цѣлъ	whole	*koyh ₁ -lo-	B	MAX	H
32	слѣдъ	trace	*sleyd ^h -	B	STR	H
33	мѣсто	place	*moy-t ^h ₂ -	B	MAX	M
36	вѣкъ	age	*weyk-	B	STR	H
37	сѣверъ	north	*keh ₁ wer-	A	MAX	M
45	грѣти	to heat	*g ^{wh} er-+-eh ₁ -	S	STR	H
66	имѣти	to have	*h ₁ em-+-eh ₁ -	S	MOD	H
67	дрѣво	tree	*dóru/drew-	C	STR	H
27	хлѣбъ	bread	PGmc *hlaibaz	loan	STR	H
71	цѣсарь	emperor	Goth. <i>kaisar</i>	loan	STR	H

Table 3: Labiality distribution of the 74-entry corpus.

Category	Count	%
MAX (5–7 distinct vowels)	12	16%
STR (3–4 distinct vowels)	27	36%
MOD (2 distinct vowels)	22	30%
LIM (limited data)	6	8%
N/A (untestable)	7	9%
Testable entries $\geq$ MOD	61/67	91%

4.4 Lability by PIE Source Group

Table 4 breaks down lability rates by PIE source group.

Table 4: Lability rates by PIE source group.

Group	Total	Testable	$\geq$ MOD	% labile
A (laryngeal $*eh_1$)	18	17	17	100%
B (diphthong)	26	25	24	96%
C (mixed/debated)	13	11	9	82%
S (stative suffix $*-eh_1-$)	5	5	5	100%
Loanwords	5	5	5	100%
Unknown	4	0	—	N/A
Total	74	67	61	91%

Both Group A and Group B show near-perfect lability rates. The slightly lower rate for Group C is expected: these entries have uncertain etymologies, so some may not derive from genuine complex nuclei. Group S confirms that the stative suffix $*-eh_1-$ behaves identically to root-level laryngeals.

4.5 Robustness: Excluding Group C

Group C contains 13 entries whose PIE etymologies are mixed, debated, or secondary. Because their inclusion could inflate or deflate the overall lability rate, we characterize the group and test whether our results depend on it.

The 13 entries fall into five categories of uncertainty:

1. **Ambiguous PIE source** (3 entries): бѣль ‘white’ ($*b^h el-$ or $*b^h oyl-$?), млѣко ‘milk’ ($*h_2melǵ-$, laryngeal at wrong position), сѣдѣти ‘to sit’ ($*sed-$ or $*seh_1d-$?). These could belong to Group A or B if the debated reconstruction were resolved.
2. **Uncertain reconstruction** (3 entries): звѣзда ‘star’ ($*ǵ^h weh_1-ǵ^h d-$?), грѣхъ ‘sin’ ($*ǵ^h rey-$?), дрѣво ‘tree’ ($*dóru$ or $*drew-$?). Multiple competing etymologies coexist.
3. **Compound or derived forms** (3 entries): невѣста ‘bride’ ($*new-wed^h-$), колѣно ‘knee’ ($*(s)kwel-$), бесѣда ‘talk’ ($*be-+sed-$). The yat position within the compound is unclear.
4. **Wanderwort or substrate** (1 entry): рѣпа ‘turnip’ ($*(s)rāps$). May not be inherited IE vocabulary at all.
5. **Unexpected or secondary yat** (3 entries): рѣчь ‘speech’ ($*rek-$, simple root without obvious complex nucleus), брѣгъ ‘shore’ ($*b^h erg^h-$, secondary), сѣкти ‘to chop’ ($*sek-$, possibly Narten ablaut).

Table 5 shows that excluding Group C does not weaken the results—it strengthens them.

The core groups—those with well-established PIE complex-nucleus etymologies—show a lability rate of 98%, with the single exception being лѣнь ‘laziness’ ($*leh_1-ni-$, LS=LIM due to scarce cognate data rather than genuine non-lability). The 82%

Table 5: Lability rates with and without Group C.

Corpus subset	Testable	$\geq$ MOD	% labile
Full corpus (A+B+C+S+Loan)	67	61	91%
Excluding Group C	56	52	93%
Core only (A+B+S+Loanwords)	52	51	98%

rate for Group C is lower precisely because some of these entries may not derive from genuine complex nuclei; indeed, the two non-labile Group C entries (звѣзда, LS = LIM; сѣкти, LS = LIM) have the lowest etymological confidence in the entire corpus. **The hypothesis is robust to the exclusion of all debated entries.**

4.6 Lability by Divergence Type

We classified each entry in both the yat corpus and the control group by its dominant divergence type (as defined in §3.3). Table 6 presents the results.

Table 6: Dominant divergence type: yat corpus vs. control group.

Dominant type	Yat corpus	Control (*e)
N (nucleus resolution)	~100%	4%
A (ablaut grades)	~0%	38%
R (regular merger)	~0%	42%
C (consonant-conditioned)	~0%	2%
Mixed / none	—	13%

The distributions are essentially non-overlapping. Every testable yat entry is dominated by Type N divergence: a single complex nucleus producing divergent reflexes across branches. By contrast, 80% of control entries are dominated by Types A and R—ablaut alternation and the regular Indo-Iranian merger, respectively. Only two control entries (4%) show Type N dominance: новъ ‘new’ (**néwos*, LS = 5) and девять ‘nine’ (**h₁néwn*, LS = 4), both containing the diphthong **ew*. Crucially, **ew* is *not* one of the yat-producing diphthongs in Slavic (**ew* → **u*/**o*, not **ě*)—but it produces the same kind of nucleus-resolution lability as **ey*/**oy*. This confirms that the lability mechanism is general to complex nuclei, while yat encodes the *specific subset* that fronted to **ě*.

This type-based analysis reveals that the raw LS comparison (3.4 vs. 2.64) actually *understates* the true divergence between yat and control words. We can quantify this by defining a **Nucleus-Resolution Score (NRS)**: for each entry, NRS = LS if the dominant divergence type is N, and NRS = 0 otherwise. Since every testable yat entry is Type N dominant, the yat corpus retains its full mean: NRS = 3.4. For the control group, only the two **ew* entries contribute non-zero NRS values (4 and 5), giving a control mean of NRS = 0.17. Table 7 summarizes the comparison.

The NRS comparison makes explicit what the raw LS obscures: the control group’s mean of 2.64 is entirely driven by ablaut alternation (Type A), the regular Indo-Iranian merger (Type R), and consonant-environment effects (Type C)—none of which reflect the position-specific nucleus-resolution mechanism that characterizes yat. When only nucleus-resolution divergence is counted, the yat-to-control ratio rises from 1.3:1 to 20:1.

Table 7: Raw Lability Score vs. Nucleus-Resolution Score.

Metric	Yat corpus	Control (*e)
Mean LS	3.4	2.64
Mean NRS	3.4	0.17

4.7 Attestation Breadth

A potential confound for the LS comparison is *attestation breadth*: if yat roots are simply better attested across IE branches (because they are older, more basic, or more widely preserved), they would have more opportunities to display vowel divergence, inflating their LS relative to less widely attested control words.

To test this, we counted the number of distinct non-Slavic IE branches (Latin/Italic, Greek, Sanskrit/Indo-Iranian, Germanic, Baltic, Celtic, Armenian, Albanian, Tocharian, Hittite/Anatolian) attesting a cognate for each root. Table 8 presents the results.

Table 8: Attestation breadth: non-Slavic IE branches per root.

	Yat corpus	Control (*e)
<i>n</i>	67	53
Mean branches	3.0	4.8
Median	3	5
Range	1–5	2–7

The confound runs in the *opposite* direction from what would be problematic. Control-group roots are attested in significantly more non-Slavic branches than yat roots (mean 4.8 vs. 3.0, $p < 0.001$, Welch’s *t*-test), yet show *lower* lability scores. This asymmetry makes sense on structural grounds: many basic control words (десять ‘ten’, ясмь ‘am’, новъ ‘new’) are among the most widely preserved PIE vocabulary, attested in 6–7 branches. Yat roots, by contrast, include loanwords (хлѣбъ, хлѣвъ, цѣсарь) with only 1–2 donor branches and several inherited roots of restricted distribution (лѣнь, лѣсъ, звѣзда).

The attestation-breadth data thus strengthens rather than undermines the LS comparison: yat words achieve their higher lability despite having *fewer* branches available to show divergence. The 29% lability premium of yat over control (LS 3.4 vs. 2.64) is, if anything, a conservative estimate.

4.8 Showcase Entries

We present six entries with full cross-IE cognate tables, selected to illustrate the strongest cases from both PIE source groups.

4.8.1 вѣдѣти ‘to know’ — PIE **weyd-* ‘to see, know’

This is arguably the strongest single entry in the corpus. The PIE root **weyd-* shows virtually every possible monophthongization pathway: Gothic preserves the diphthong as *ai*, Latin and English resolve to *i/ī*, Sanskrit has *e* (from the o-grade **woyd-*), and Greek shows both *i* and *ei*. The Slavic yat reflexes (e/i/je) recapitulate this full range within a single language family. **LS = MAX.**

Table 9: Cross-IE reflexes of **weyd-* at the yat position.

Branch	Cognate	Vowel
Slavic Ekavian	<i>vedeti</i>	e
Slavic Ikavian	<i>viditi</i>	i
Slavic Ijekavian	<i>vidjeti</i>	i/je
Latin	<i>vidēre, vīdī</i>	i, ī
Greek	<i>ideîn, eîdos</i>	i, ei
Sanskrit	<i>vēda</i>	e
English	<i>wit, wise, wizard</i>	i
Gothic	<i>wait</i>	ai

4.8.2 цѣна ‘price, penalty’ — PIE **k^woynéh₂* ‘payment’

Table 10: Cross-IE reflexes of **k^woynéh₂* at the yat position.

Branch	Cognate	Vowel
Slavic Ekavian	<i>cena</i>	e
Slavic Ikavian	<i>cina</i>	i
Slavic Ijekavian	<i>cijena</i>	ije
Greek	<i>poinē</i> (penalty)	oi
Latin	<i>poena</i> (punishment)	oe
Lithuanian	<i>káina</i>	ai
Avestan	<i>kaēnā</i>	ae
Old Irish	<i>cín</i> (penalty)	í

Greek *poinē* preserves the original diphthong **oy* as *oi*. Latin keeps it as *oe*. Lithuanian and Avestan show *ai/ae*. Old Irish monophthongizes to *ī*. The spread *oi* – *oe* – *ai* – *ae* – *ī* – *e/i/ije* across six branches is maximal. English *pain* and *penal* are Latin loans from this root. **LS = MAX.**

4.8.3 вѣтръ ‘wind’ — PIE **h₂weh₁-nt-* ‘blowing’

Table 11: Cross-IE reflexes of **h₂weh₁-nt-* at the yat position.

Branch	Cognate	Vowel
Slavic Ekavian	<i>vetar</i>	e
Slavic Ikavian	<i>vitar</i>	i
Slavic Ijekavian	<i>vjetar</i>	je
Latin	<i>ventus</i>	e
English	<i>wind</i>	i
Gothic	<i>winds</i>	i
Sanskrit	<i>vāta</i>	ā
Lithuanian	<i>vėtra</i>	ė

Five branches produce five different vowels from the same root: Latin *e*, English/Gothic *i*, Sanskrit *ā*, Lithuanian *ė*, and Slavic *e/i/je*. This is a textbook example of laryngeal-derived labiality. **LS = MAX.**

4.8.4 цѣлъ ‘whole, healthy’ — PIE **koyh₁-lo-* ‘whole, unharmed’

Table 12: Cross-IE reflexes of **koyh₁-lo-* at the yat position.

Branch	Cognate	Vowel
Slavic Ekavian	<i>ceo</i>	e
Slavic Ikavian	<i>cil</i>	i
Slavic Ijekavian	<i>cijel</i>	ije
Gothic	<i>hails</i> (whole, healthy)	ai
Old English	<i>hāl</i> (→ <i>whole</i> , <i>hale</i>)	ā
German	<i>heil</i>	ei
Welsh	<i>coel</i> (omen)	oe

Gothic *hails* preserves the PIE diphthong **oy* as *ai*. Old English monophthongizes to *ā* (whence Modern English *whole* and *hale*). German preserves *ei*. Welsh shows *oe*. A single PIE diphthong produces five distinct vowel outcomes in five branches. English *whole*, *hale*, *heal*, and *health* all descend from this root. **LS = MAX.**

4.8.5 снѣгъ ‘snow’ — PIE **sneyg^{wh}-* ‘to snow’

Table 13: Cross-IE reflexes of **sneyg^{wh}-* at the yat position.

Branch	Cognate	Vowel
Slavic Ekavian	<i>sneg</i>	e
Slavic Ikavian	<i>snig</i>	i
Slavic Ijekavian	<i>snijeg</i>	ije
Latin	<i>nix</i> , <i>niv-is</i>	i
Old English	<i>snāw</i> (→ <i>snow</i>)	ā
Gothic	<i>snaiws</i>	ai
Lithuanian	<i>sniėgas</i>	ie
Sanskrit	<i>snihyati</i> (gets wet)	i

Gothic *snaiws* preserves the diphthong. Latin and Sanskrit resolve to *i*. Old English develops *ā* (from the o-grade **snoyg-*). Lithuanian has *ie*. The mapping to Slavic *e/i/je* is strikingly parallel. **LS = MAX.**

4.8.6 млѣко ‘milk’ — PIE **h₂melǵ-* ‘to milk, stroke’

This entry illustrates the hypothesis with particular clarity: Latin *lac* with *a* parallels Bulgarian *mljako* with *ja*; English *milk* with *i* parallels Ikavian *mliko*; Greek *amélgō* with *e* parallels Ekavian *mleko*. All four Slavic reflexes find their mirrors in other IE branches. **LS = MAX.**

Table 14: Cross-IE reflexes of **h₂melǵ-* at the yat position.

Branch	Cognate	Vowel
Slavic Ekavian	<i>mleko</i>	e
Slavic Ikavian	<i>mliko</i>	i
Slavic Ijekavian	<i>mlijeko</i>	ije
Bulgarian	<i>mljako</i>	ja
Latin	<i>lac, lact-is</i>	a
Greek	<i>amélgō</i> (to milk)	e
English	<i>milk</i>	i
Old English	<i>meolc</i>	eo

5 The Loanword Test

5.1 Prediction

If yat encodes a phonological class (the “complex nucleus slot”) rather than a specific sound, then early loanwords into Proto-Slavic should obey the following rule:

Prediction: All early loanwords where the donor language has a diphthong **ai* or a long front vowel **ē* should receive yat (ѣ) in Slavic. Loanwords with simple short vowels, back diphthongs (**au*, **eu*), or long back vowels should **not** receive yat.

This prediction is independently testable because the donor forms are *directly attested* in Gothic, Old High German, and other early sources—we do not need to reconstruct them from Slavic evidence.

5.2 Loanwords with Yat

Table 15 lists all early loanwords into Proto-Slavic that bear yat.

Table 15: Early loanwords into Proto-Slavic with yat: source vowel analysis.

#	Slavic form	Meaning	Donor form	Source vowel	Yat?
1	<i>*xlěbŭ</i>	bread	PGmc <i>*hlaibaz</i>	diphthong <i>*ai</i>	Yes
2	<i>*cěsarŭ</i>	emperor	Goth. <i>kaisar</i>	diphthong <i>*ai</i>	Yes
3	<i>*xlěvŭ</i>	cowshed	PGmc <i>*hlaiw-</i>	diphthong <i>*ai</i>	Yes
4	<i>*lěkŭ</i>	medicine	PGmc <i>*lēkiz</i>	long <i>*ē</i>	Yes
5	<i>*strěla</i>	arrow	WGmc <i>*strēla</i>	long <i>*ē</i>	Yes [†]
6	<i>*rědiky</i>	radish	Lat. <i>rādīx</i> via WGmc	long <i>*ā</i>	Yes
7	<i>*pěnědžŭ</i>	penny	PGmc <i>*panningaz</i>	short <i>*a[‡]</i>	Yes

[†] Direction of borrowing debated; may be shared PIE inheritance.

[‡] Edge case: source may be WGmc i-umlaut **penning* with tense/long *e*.

Six out of seven yat-bearing loanwords have a clear diphthong **ai* or long front vowel in the source. The seventh (**pěnědžŭ*) has a plausible West Germanic i-umlaut explanation.

5.3 Loanwords without Yat

We examined 38 additional early loanwords into Proto-Slavic that do *not* bear yat. Their source vowels distribute as follows:

- **Diphthong **au*** (5 entries): **kupiti* ‘to buy’ ← Gothic *kaupjan*; **kusiti* ‘to taste’ ← Gothic *kausjan*; **nuta* ‘cattle’ ← PGmc **nautq*; **lukŭ* ‘leek’ ← PGmc **lauka*; **gunŭja* ‘cloak’ ← Iranian **gaunyā*. All give Slavic **u*, never yat.
- **Diphthong **eu*** (1 entry): **bljudo* ‘dish’ ← PGmc **beudq*. Gives **u*.
- **Diphthong **ei* (monophthongized to **ī*)** (2 entries): **vino* ‘wine’ ← Gothic *wein* [i:]; **lixva* ‘usury’ ← Gothic *leihwa* [i:]. Both give Slavic **i*, not yat.
- **Long back vowels** (5 entries): long **ō* → **u*; long **ū* → **y*. Never yat.
- **Long **ī*** (1 entry): **vitędzŭ* ‘knight’ ← PGmc **wikingaz*. Gives **i*, not yat.
- **Short vowels** (17 entries): **oslŭ* ‘donkey’, **kotlŭ* ‘cauldron’, **gradŭ* ‘town’, **skotŭ* ‘cattle’, etc. All short vowels. **None** have yat.

All 38 non-yat loanwords conform to the prediction: zero false positives.

The pattern extends beyond Germanic. We surveyed 10 confirmed Iranian loanwords in Proto-Slavic (**bogŭ* ‘god’ ← **baga*-; **toporŭ* ‘axe’ ← **tapara*-; **xvala* ‘praise’ ← **xvarənah*-; etc.): all have simple back or short vowels, and **none** bear yat. The Iranian case is phonologically predicted: by the Scythian-Sarmatian contact period, Iranian diphthongs had already monophthongized, unlike Gothic which preserved live **ai*. Even Iranian **au* follows the Germanic pattern: **gunŭja* ‘cloak’ from Iranian **gaunyā* gives Slavic **u*, just as **kupiti* from Gothic *kaupjan* does.

Similarly, 11 Greek ecclesiastical loanwords borrowed after iotacism (ангелъ, монахъ, епископъ, евангелие, ересь, etc.) uniformly lack yat, despite two of them (ересь ← *aἰρεσις*, еретикъ ← *aἰρετικός*) deriving from historical *ai*—by the 9th century a simple [e]. Six direct Latin loans with simple vowels (**poganŭ* ‘pagan’, **mŭša* ‘mass’, **kolęda* ‘carols’, etc.) likewise lack yat.

Including these non-Germanic sources raises the total non-yat loanword count from 38 to approximately 55, with no change to the zero-false-positive rate.

5.4 Contingency Table and Significance

Table 16: Contingency table for loanword vowel mapping to yat.

	Source: <i>*ai</i> /long <i>*ē</i> /long <i>*ā</i>	Source: other vowel
Slavic has yat	6 (+1 edge case)	0
Slavic lacks yat	0 (+2 post-iotacism <i>*ai</i>)	~55

Including the expanded non-Germanic data, the total corpus is approximately 62 loanwords. **The original Germanic-only accuracy stands at $44/45 = 97.8\%$; the expanded corpus shows the same zero-false-positive rate across all source languages.** Fisher’s exact test on the 2×2 table yields $p < 0.0001$. The two post-iotacism Greek *ai* words (ересь, еретикъ) are counted in the “other vowel” column

because the diphthong had already monophthongized to [e] at the time of borrowing—a classification confirmed by the ересъ/ѡѣсаръ contrast discussed below.

5.5 The Critical Contrasts

Three contrasts are particularly informative:

Gothic *wein* vs. Gothic *hlaifs*. Gothic *wein* [wi:n] had already monophthongized **ei* to [i:] before the Slavic borrowing. Slavs received a monophthong and mapped it to **i*: hence **vino* ‘wine’, with no yat. Gothic *hlaifs* retained the diphthong **ai* as [æ:] or [ai]. Slavs mapped this live diphthong to yat: hence **xlěbŭ* ‘bread’, with yat. Slavic speakers distinguished between a monophthongized former diphthong (→ no yat) and a living diphthong or long front vowel (→ yat). This demonstrates remarkable phonological sensitivity.

Diphthong **ai* vs. diphthong **au*. Five loanwords with **au* all give Slavic **u*, never yat. Three loanwords with **ai* all give yat. This means yat was not “the diphthong slot” in general—it was specifically the *front diphthong* / *long front vowel* slot. Back diphthongs had their own resolution path (**au* → **u*), exactly as expected from the PIE-level phonology.

Greek ересъ vs. Gothic ѡѣсаръ: the same diphthong, opposite outcomes. Both words ultimately derive from the same historical diphthong **ai*: Latin *Caesar* (ae = [ai]) entered Slavic via Gothic *kaisar*, where **ai* was still a live diphthong; Greek *αἰρεσις* (*hairesis*) entered Slavic directly from Byzantine Greek, where *ai* had already iotacized to [e] by the 9th century. The results are opposite: ѡѣсаръ has yat, ересъ does not.

This contrast is decisive because it controls for etymology: the source diphthong is identical (**ai*), but the *phonetic substance at the time of borrowing* differs. Slavic speakers assigned yat when they heard a live diphthong or long open-front vowel, and withheld it when they heard a simple [e]—regardless of the historical origin of that vowel. This proves that yat assignment was a phonological process responding to acoustic input, not a learned convention or orthographic inheritance.

5.6 Breaking the Circularity

A potential objection to our inherited-vocabulary results is circularity: PIE complex nuclei (**eh₁*, **ey*, etc.) are partly reconstructed *from* the cross-branch vowel divergence we are measuring. If the lability pattern is built into the reconstruction method, we may be rediscovering our own assumptions.

We address this objection on three fronts.

Loanwords break the circle. The loanword test (§5) provides the strongest counterargument. The Germanic donor forms—Gothic *hlaifs*, *wein*, *kaisar*, *kaupjan*—are *independently attested* in written sources. We know the source vowel quality without relying on Slavic evidence or PIE reconstruction. The fact that Proto-Slavic speakers mapped these known vowels to yat in a perfectly systematic way (98% accuracy) proves that yat was a living phonological category, not an artifact of the comparative method.

The prediction is asymmetric. Even if PIE reconstructions were partly circular, the hypothesis makes a *specific directional prediction*: that Slavic should have encoded the complex-nucleus class with a dedicated grapheme, while non-complex-nucleus roots should lack that grapheme. The control group of 53 non-yat **e* words tests this prediction directly. The fact that non-yat words achieve their (lower) lability through qualitatively different mechanisms (Types A and R rather than Type N; §4.6) is not predicted by circularity alone—it requires a real phonological distinction between the two classes.

Consonantal and morphological evidence. Many of the PIE etymologies in our corpus are supported by *non-vowel* evidence: regular consonantal correspondences (e.g., **k^w* → Latin *p*-, Greek *t-/p*-, Slavic *c*- in цѣна), root structure and ablaut patterns, and independent morphological analysis. The vowel-quality divergence we measure at the yat position is therefore not the sole basis for the PIE reconstruction—it is one consequence among several. We test this argument formally in the following subsection.

5.7 The Non-Circular Subset

To quantify how much of our result survives when the circularity objection is taken to its logical extreme, we identified a **non-circular subset**: entries where the PIE root assignment is supported by non-vowel evidence sufficient for a skeptic to accept the reconstruction even if the vowel at the yat position were unknown. We required at least one of:

- **Consonantal correspondence chain** — regular stop-series or cluster correspondences across branches (e.g., **d^h* → Latin *f*-, Greek *th*-, Germanic *d*-, Slavic *d*- for дѣло; **sn*- cluster for снѣгъ; *m-l-g* skeleton for млѣко);
- **Morphological structure** — identifiable PIE suffix (**-men*-, **-nt*-, **-ro*-, **-lo*-) or derivational pattern;
- **Transparent compound** — independently supported constituent roots (e.g., медвѣдь = **medhu* + **h₁ed*- “honey-eater”; гнѣздо = **ni-sd*- “sitting-down-place”);
- **Loanword attestation** — donor form directly attested in written sources.

Of the 67 testable entries, 41 meet this standard (see supplementary file for full tagging). The remaining 26 have thin consonantal frames (single-consonant roots like **mey*-), uncertain etymologies, or limited attestation breadth.

Table 17: Lability in the non-circular subset vs. full corpus.

	Non-circular (41)	Full testable (67)
Labile ($\geq$ MOD)	41	61
Lability rate	100%	91%
MAX entries	9 (22%)	12 (18%)
STR entries	19 (46%)	~27 (40%)
MOD entries	13 (32%)	22 (33%)
LIM entries	0 (0%)	6 (9%)

The result is unambiguous: restricting to entries where the PIE reconstruction stands on non-vowel evidence *raises* the lability rate from 91% to **100%**. All six LIM entries—the only entries with lability below MOD in the full corpus—fall outside the non-circular subset, because they have both uncertain etymologies and limited attestation breadth. This is not a coincidence: entries with strong consonantal evidence tend to be the same entries with cognates in multiple branches, providing ample material for the vowel divergence to manifest.

The non-circular subset spans both PIE source groups (14 Group A, 15 Group B, 3 Group C, 3 Group S, 3 loanwords) and all four evidence types. Representative entries include labiovelar chains (цѣна, снѣгъ, грѣти), aspiration chains (дѣло, дѣва, звѣрь), morphologically transparent compounds (медвѣдь, гнѣздо, обѣдь), and loanwords with independently attested donor forms (хлѣбъ, цѣсарь, хлѣвъ). In every case, the consonant frame identifies the PIE root; the vowel at the yat position is a *consequence* of the reconstruction, not its input.

6 The Dialect Exception Test

6.1 Prediction

If Group A (laryngeal **eh₁*) and Group B (diphthong **ey/ *oy/ *ay*) words carry different “resolution pressures,” then dialect exceptions—words where a given dialect deviates from its expected yat reflex—might cluster by PIE source group. For example, an Ekavian dialect showing occasional Ikavisms might show those Ikavisms predominantly in Group B words (or predominantly in Group A).

6.2 Evidence from Dialect Exceptions

We catalogued yat reflex exceptions from five sources: (1) standard Ekavian Serbian grammatical Ikavisms (*nisam*, comparative *-iji*, instrumental *-im*); (2) dialectal Ikavisms in Šumadija-Vojvodina Ekavian; (3) Ijekavian → /i/ exceptions before /o/ from vocalized /l/; (4) Zeta-Raška Ijekavian → /i/ before /j/; and (5) Ijekavian → /e/ after /r/ + consonant cluster.

Table 18: Distribution of yat reflex exceptions by PIE source group.

Exception type	Grp A	Grp B	Mixed	Dominant factor
Standard Ek. Ikavisms	1?	2	1	Morphological
Dialectal Ek. Ikavisms	2	2	1	Phonological (/j/)
Ijek. → /i/ before /o/	1	1–2	3	Phonological (/o/ < /l/)
Ijek. → /i/ before /j/	1?	0	1	Phonological (palatal)
Ijek. → /e/ after r+C	0–1	1	1	Phonological (cluster)

Total clear exceptions: Group A $\approx$ 5–6, Group B $\approx$ 6–7. Given 10 Group A and 12 Group B words in the original testable corpus, the expected ratio is $\sim$ 45% / 55%. The observed ratio is $\sim$ 45% / 55%. **No statistically meaningful clustering by PIE source group is observed.**

6.3 Jakubinskij’s Law

The strongest test comes from Middle Čakavian dialects, where Jakubinskij (1925) and Meyer (1926) documented a systematic yat split: $*\check{e} \rightarrow /e/$ before a dental {d, t, s, z, n, l, r} followed by a back vowel {a, o, u, ŭ}; $*\check{e} \rightarrow /i/$ elsewhere. We verified this law against our classified corpus: it cuts across both Group A and Group B with equal force. The same root can produce /e/ in one morphological form and /i/ in another depending solely on the following segment. PIE etymology is completely irrelevant to this split.

6.4 Interpretation

Prediction 8 is not confirmed. The known yat reflex exceptions are overwhelmingly conditioned by synchronic phonological environment (dental vs. non-dental consonant, back vs. front vowel, palatal context, r + cluster) and by morphological category (high-frequency grammatical morphemes). PIE etymological source plays no detectable role.

The most likely explanation is that the two PIE sources of yat had *fully merged* in Proto-Slavic before any dialect-internal yat splitting began. By the time Čakavian, Štokavian, and other dialects started resolving yat differently (roughly 12th–14th centuries CE), the laryngeal vs. diphthong distinction was 3,000+ years old and had left no phonetic trace. Proto-Slavic $*\check{e}$ was a single phoneme regardless of its PIE origin.

This negative result does not weaken the core hypothesis. It clarifies a chronological boundary:

PIE complex nucleus $\rightarrow$ **cross-IE lability** (each branch resolves differently)
 $\rightarrow$ Proto-Slavic $*\check{e}$ merger (Groups A and B become one phoneme)
 $\rightarrow$ **intra-Slavic splits** (conditioned by synchronic phonology)

Yat captures the *first* level of lability (cross-IE), not the second (intra-Slavic).

7 Discussion

7.1 The Four Slavic Reflexes as Four IE Resolution Strategies

The most striking finding is that the four modern Slavic yat reflexes are not arbitrary—they recapitulate the four main resolution strategies that other IE branches independently adopted for the same complex nuclei:

Table 19: Correspondence between Slavic yat reflexes and cross-IE resolution strategies.

Slavic reflex	IE parallel	Strategy
Ekavian /e/	Latin <i>sedeo</i> , Greek <i>hédra</i> , Sanskrit <i>śvéta</i>	e-grade preservation
Ikavian /i/	Latin <i>vidēre</i> , English <i>sit/wit/milk</i>	i-monophthongization
Ijekavian /ije/	Lithuanian <i>sniēgas</i> , Avestan <i>kaēnā</i>	diphthong preservation
Bulgarian /ja/	Latin <i>lac</i> , OHG <i>sāmo</i> , Lithuanian <i>bālas</i>	a-coloring

Each Slavic dialect chose one of the same resolution paths that entire IE branches chose millennia earlier. The Slavic-internal divergence is, in this sense, a recapitulation of the IE-level divergence—driven by the same phonological instability inherent in the PIE complex nucleus.

7.2 Yat as a “Written Laryngeal”

In PIE reconstruction, laryngeals ($*h_1$, $*h_2$, $*h_3$) are abstract segments posited to explain otherwise inexplicable vowel alternations across IE branches. They were lost as consonants but left traces as vowel coloring and lengthening. Yat behaves structurally like a **visible laryngeal-trace marker**:

1. It marks a position where a PIE complex nucleus (V + laryngeal, or diphthong) existed.
2. At that position, each daughter language resolves the vowel differently.
3. The symbol Ъ does not specify *which* resolution—it marks the *slot*.

At minimum, Cyril encoded a phonetic distinction present in his native Thessalonian Slavic dialect. Proto-Slavic $*ě$ was a distinct phoneme—probably [æ:] or [ɛ:], as constrained by the loanword evidence (§7.4)—and Cyril, as a careful phonologist designing an alphabet, gave it its own letter. He did not need to know *why* this vowel class existed or how it related to PIE; he needed only to hear *that* it was distinct from /e/. The structural correspondence between yat and PIE complex nuclei is then a consequence of phonological inheritance, not of Cyril’s analytical intentions.

However, two considerations suggest that yat may have been more than an ordinary phoneme even in Cyril’s time.

The acoustic breadth of yat. The loanword evidence (§5) shows that Proto-Slavic speakers mapped *phonetically distinct* source sounds to yat: Germanic $*ai$ [æ:] or [ai], long $*ē$ [e:], and Latin *ae*—sounds spanning from open-front to mid-front. Meanwhile, Germanic [i:] was categorically rejected (mapped to /i/, not yat), as were all back vowels. This means the yat category occupied an unusually *broad* acoustic range within the front vowel space, bounded sharply above by [i:] and below/behind by back vowels but covering a span from approximately [æ:] to [e:]. A phoneme with this kind of breadth is inherently more susceptible to allophonic variation: different communities could anchor their realization at different points within the range—some closer to [æ:], others to [ɛ:], others to [e:]—while still producing “the same phoneme.” This breadth may be a direct phonetic legacy of the PIE complex nucleus, whose multiple resolution pathways gave the category its wide acoustic footprint in the first place.

The Balkan context. The South Slavic dialect continuum is unique among Slavic-speaking regions in preserving *all four* major yat reflexes—Ekavian /e/, Ikavian /i/, Ijekavian /ije/, and Bulgarian /ja/—in close geographic proximity. In no other Slavic region do all four coexist: Russian settled uniformly on /e/, Ukrainian on /i/, Czech on /je ~ e/, Polish merged yat entirely. Only in the Balkans, and particularly in the Štokavian-Čakavian continuum, do speakers of all three reflexes live within kilometers of each other—sometimes within the same city. Cyril, based in 9th-century Thessaloniki, a cosmopolitan city surrounded by Slavic-speaking populations of diverse tribal and dialectal origins, was situated at the epicenter of this eventual fragmentation.

The standard chronology places the full yat split at the 12th–14th centuries, well after Cyril. Our dialect exception test (§6) shows that the splits are conditioned by synchronic phonology (dental environments, palatal contexts), not by PIE etymology—suggesting they represent a later phonological reorganization. Early OCS manuscripts show no

spelling hesitation at yat positions, which argues against perceptible dialectal variation in the 9th century. The combination of an acoustically broad phonemic category and the Balkan linguistic environment may explain why **ě* was encoded with a dedicated grapheme despite showing no overt dialectal splits in Cyril’s time.

Regardless of what Cyril perceived, the result is remarkable from a modern perspective. Yat may be the only place in the entire IE written record where a post-laryngeal phonological pattern was captured as a synchronic, usable grapheme—not reconstructed backward by scholars, but written down by a literate speaker who heard the reflex of a PIE complex nucleus as a living sound. His alphabet preserves the pattern whether or not he understood it.

7.3 Grammatical Yat: The Stative Suffix

Five verbs in our corpus bear yat in the stative verbal suffix **-eh₁-* rather than in the root: грѣти ‘to heat’, дрѣмати ‘to doze’, имѣти ‘to have’, болѣти ‘to be sick’, and терпѣти ‘to endure’. The suffix **-eh₁-* is itself a laryngeal-derived complex nucleus, so it correctly predicts cross-IE lability—but the lability appears at the *suffix* position, not the root. This distinction between *lexical yat* (in the root) and *grammatical yat* (in derivational morphology) has not been systematically noted in the literature but is a natural consequence of the complex-nucleus hypothesis.

7.4 Chronological Implications

The loanword data allow us to establish a chronological bracket for yat as a productive phonological category. Germanic loanwords datable to the 4th–6th centuries CE—**xlěbŭ* from **hlaibaz*, **cēsari* from *kaisar*, **xlěvŭ* from **hlaiw*—all show that Proto-Slavic speakers could hear a Germanic **ai* and correctly assign it to the yat class. This means **ě* was still a *living, distinct phonological category* during the Gothic contact period, not a historical relic.

At the same time, the *wein/hlaifs* contrast demonstrates that Slavic speakers were sensitive to the *status* of the source diphthong: Gothic **ei*, which had already monophthongized to [i:] before the borrowing, was mapped to Slavic **i* (**vino*), not yat. Only diphthongs that were still phonetically “live” at the time of borrowing received yat. This allows us to constrain yat’s phonetic value during the loanword period:

Table 20: Loanword constraints on the phonetic value of Proto-Slavic **ě*.

Source vowel	Example	Slavic result	Implication
Gmc <i>*ai</i> [æ:] or [ai]	<i>hlaifs</i>	→ yat	<i>*ě</i> compatible with [æ:]
Gmc <i>*ē</i> [e:]	<i>*lēkiz</i>	→ yat	<i>*ě</i> compatible with [e:]
Gmc <i>*ī</i> [i:]	<i>wein</i>	→ <i>*i</i> , not yat	<i>*ě</i> was not [i:]
Gmc <i>*au</i> [o:]	<i>kaupjan</i>	→ <i>*u</i> , not yat	<i>*ě</i> was not a back vowel

These constraints converge on a phonetic value of approximately [æ:] or [ɛ:] for Proto-Slavic **ě* in the 4th–6th centuries—a long mid-to-open front vowel, close enough to Germanic **ai*/**ē* to be identified with them, but distinct from **ī* and **au*.

Three additional considerations narrow the range further, favoring [æ:] over [ɛ:].

Reverse borrowings: Romanian *ea*. When Slavic words containing **ě* were borrowed *into* neighboring languages, the receiving language had to substitute its own vowel for **ě*. In Romanian, which has a massive Slavic superstrate layer dating from the 6th century onward, the pattern is systematic: Slavic **ě* → Romanian *ea* [ɛa] (*neveastă* ← **nevěsta* ‘bride’; *mreață* ← **mrěža* ‘net’; *treabă* ← **trěba* ‘need’), while Slavic plain **e* → Romanian *e* (*nevod* ← **nevodŭ* ‘dragnet’; *peșteră* ← **peštera* ‘cave’). The Romanian diphthong *ea* [ɛa] spans the acoustic range from [e] to [a], with its midpoint at approximately [æ]. If **ě* were [ɛ:], Romanian would more likely render it as simple *e* [e], its closest monophthong; the diphthongal rendering suggests the source was lower than [ɛ]—i.e., closer to [æ:]. Old Romanian Cyrillic orthography confirms this identification: the yat letter Ъ was used specifically to write the *ea* diphthong.

The Bulgarian yat split. Modern Bulgarian shows an environmentally conditioned split of the former yat: [a] (with preceding palatalization) before back vowels and word-finally (*бял* [bjal], *мляко* [mljako], *хляб* [xljap]), but [ɛ] before front vowels (*белѝ* [bɛli], *хлебѝ* [xlɛbar]). This split is naturally explained if the starting point was [æ:], which is equidistant from both endpoints on the vowel chart: lowering to [a] and raising to [ɛ] are both one-step processes. From [ɛ:], the shift to [a] requires two steps—a much harder development to motivate phonetically. The letter name *ѡмъ* itself contains iotated yat (written ѡтъ), so its modern Bulgarian pronunciation [jat] directly attests the [æ:] → [a] development.

The [i] exclusion gap. If **ě* were [ɛ:], the distance from Gothic [i] to **ě* would be one step on the vowel chart—yet Slavic categorically rejected [i:] as a yat source (**vino*, **lixva*). If **ě* were [æ:], the distance from [i:] is two steps, making the categorical exclusion more natural. The same logic explains why post-iotacism Greek [e], though close to [ɛ:], was not mapped to yat: from [æ:], Greek [e] differs in both quality *and* length, providing a double barrier.

These three arguments—Romanian *ea*, the Bulgarian split, and the [i:] exclusion—jointly favor [æ:] as the specific phonetic value of Proto-Slavic **ě*. This aligns with Lunt’s reconstruction rather than Shevelov’s higher [ɛ:].

Combining this with the dialect exception data, we can reconstruct the following chronology:

Table 21: Chronology of yat from PIE to the modern dialect splits.

Period	Approximate dates	Yat status
Early Proto-Slavic	~1000–500 BCE	<i>*eh₁</i> and <i>*ey</i> / <i>*oy</i> merging → <i>*ě</i>
Late Proto-Slavic	~500 BCE – 500 CE	single phoneme [æ:]
Loanword period	~300–600 CE	still a live category (loanword test)
OCS creation	~863 CE	Cyril encodes <i>*ě</i> as Ъ
Pre-split period	~900–1200 CE	allophonic variation emerging
Dialect splits	~1200+ CE	e / i / ije / ja

The critical observation is that the 3,000+ year gap between the PIE complex-nucleus stage (~3000 BCE) and the intra-Slavic dialect splits (~1200 CE) explains why no etymological trace of the laryngeal/diphthong distinction survives in the mod-

ern reflexes. The two PIE sources had been fully merged for millennia before the new round of splitting began.

7.5 Limitations and Future Work

Several limitations should be acknowledged. First, while the raw Lability Score conflates qualitatively different types of variation, we have addressed this through both the divergence-type classification and the Nucleus-Resolution Score (§4.6); our NRS is binary (Type N dominant or not) and could be refined into a continuous measure that apportions each entry’s LS among contributing divergence types. Second, we have shown (§4.5) that excluding Group C (13 debated entries) strengthens the lability rate from 91% to 93%; the core groups reach 98%. Nevertheless, a thorough reassessment of Group C entries—particularly рѣчь ‘speech’ and брѣгъ ‘shore’, which have high-confidence cognates but unexpected yat—could either expand or contract the scope of the hypothesis. Third, we have shown (§4.7) that the attestation-breadth confound runs in the wrong direction—control roots are attested in more branches than yat roots—but our branch counts rely on standard reference works and may undercount poorly documented branches. Fourth, the “written laryngeal” framing (§7) describes the structural *result* of Cyril’s graphemic choice; while we argue there that the loanword breadth and Balkan context make yat an unusually unstable phonemic category, the degree to which Cyril perceived this instability remains an open empirical question that manuscript variation studies might help resolve. Fifth, the non-circular subset (§5.7) achieves a 100% lability rate across 41 entries, but the tagging of “sufficient non-vowel evidence” involves judgment calls; a more conservative tagger might retain fewer entries, and a more generous one might include some of our marginal cases.

Future work should pursue several directions: (1) refining the binary NRS into a continuous measure that apportions each entry’s LS among contributing divergence types; (2) testing whether the earliest OCS manuscripts (Codex Zographensis, Codex Marianus, Codex Suprasliensis) apply yat consistently to the complex-nucleus class, or whether scribal usage evolved; (3) psycholinguistic experiments measuring whether modern speakers of Ekavian, Ikavian, and Ijekavian dialects show different processing costs for yat-class vs. non-yat words; (4) extending the analysis to other “anomalous” Slavic vowels (e.g., the jers ѣ and ѥ) to test whether similar PIE-level patterns underlie their behavior; and (5) testing whether other IE branches show analogous residual instability at the same positions—for example, whether the Latin $\bar{e}/\bar{i}$ and Germanic $ai/\bar{a}$ alternations at yat-equivalent positions show dialect-internal variation within those branches.

8 Conclusion

We have presented three lines of evidence regarding the phonological nature of yat (Ѣ):

1. **Cross-IE lability is systematic.** Of 67 testable yat-bearing words, 91% show significant vowel divergence across IE branches at the yat position, with a mean Lability Score of 3.4 compared with 2.64 for a 53-word non-yat control group ($p < 0.001$). The difference is qualitative as well as quantitative: when only nucleus-resolution divergence is counted (NRS), the yat-to-control ratio rises from 1.3:1 to 20:1, because yat words derive their lability from the resolution of a single

complex nucleus while control words achieve theirs through ablaut alternation, regular mergers, or consonant effects. Both laryngeal-derived (Group A, 100%) and diphthongal (Group B, 96%) yat words show the pattern.

2. **Loanwords confirm the phonological class.** Across 45 early Germanic loanwords into Proto-Slavic, the mapping of source vowel quality to yat is 98% accurate: **ai* diphthongs and long **ē* → yat; all other vowel types → not yat. The pattern extends to Iranian (0/10 with yat, as predicted by their simple vowel systems), Latin, and post-iotacism Greek loans. The ересѣ/ѣресѣ contrast—same historical diphthong **ai*, opposite Slavic outcomes depending on whether the diphthong was still live at borrowing—demonstrates that yat assignment responded to phonological substance, not etymology. Reverse borrowings into Romanian (yat → *ea* [ɛa]) and Hungarian (yat → *é* [e:]) further constrain the phonetic value to [æ:], consistent with Lunt’s reconstruction.
3. **Dialect exceptions do not correlate with PIE source.** The two PIE source groups (laryngeal and diphthong) had fully merged into a single Proto-Slavic phoneme before intra-Slavic divergence began. Modern dialect exceptions are conditioned by synchronic phonology, not by PIE etymology.

Together, these findings support the interpretation of yat as a graphemic marker of PIE-level vowel instability—a position in the root where the phonological nucleus was complex, giving each IE branch (and, subsequently, each Slavic dialect) the material to resolve the vowel differently. Yat does not encode a sound; it encodes a *site of divergence*.

A Full Corpus

Table 22 presents all 74 entries in the corpus. For full cognate tables, see the supplementary materials.

Table 22: Complete yat corpus: 74 entries with PIE source and lability classification.

#	OCS word	Meaning	PIE root	Grp	LS	Conf
1	дѣло	deed	*d ^h eh ₁ -	A	MOD	H
2	сѣмя	seed	*seh ₁ -men-	A	STR	H
3	мѣра	measure	*meh ₁ -	A	STR	H
4	вѣра	faith	*weh ₁ -ro-	A	STR	H
5	вѣтръ	wind	*h ₂ weh ₁ -nt-	A	MAX	H
6	мѣсяць	month	*meh ₁ -ns-	A	MOD	H
7	лѣнь	laziness	*leh ₁ -ni-	A	LIM	M
8	вѣдѣти	to know	*weyd-	B	MAX	H
9	снѣгъ	snow	*sneyg ^{wh} -	B	MAX	H
10	цвѣтъ	flower	*k ^l weyt-	B	STR	H
11	бѣлъ	white	*b ^h el-/b ^h oyl-?	C	STR	M
12	цѣна	price	*k ^w oyne ^h ₂	B	MAX	H
13	бѣда	misfortune	*b ^h eyd ^h -	B	STR	H
14	гнѣздо	nest	*ni-sd-ó-	B	STR	H
15	мѣна	exchange	*mey-	B	STR	M
16	рѣка	river	*h ₁ reyH-	B	MOD	M
17	млѣко	milk	*h ₂ melǵ-	C	MAX	M
18	сѣдѣти	to sit	*sed-/seh ₁ d-?	C	MAX	M
19	лѣто	summer	*leh ₁ -to?	A?	LIM	L
20	стѣна	wall	*steyH-no-	B	MOD	M
21	пѣна	foam	*(s)poyH-no-	B	STR	M
22	цѣлъ	whole	*koyh ₁ -lo-	B	MAX	H
23	дѣва	maiden	*d ^h eh ₁ (i)-	A	STR	H
24	звѣрь	beast	*ǵ ^h weh ₁ r-	A	MOD	H
25	стрѣла	arrow	*streh ₁ l- [†]	A	STR	H

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#	OCS word	Meaning	PIE root	Grp	LS	Conf
26	вѣнецъ	wreath	*woyh ₁ -no-	B	STR	M
27	хлѣбъ	bread	PGmc *hlaibaz	loan	STR	H
28	лѣсъ	forest	*leyH-s-?	B?	LIM	L
29	звѣзда	star	*g ^h weh ₁ -g ^h d-?	C	LIM	L
30	сѣно	hay	*koinom	B	MOD	H
31	рѣпа	turnip	*(s)rāps (Wand.)	C	MAX	M
32	слѣдъ	trace	*sleyd ^h -	B	STR	H
33	мѣсто	place	*moy-t ^h ₂ -	B	MAX	M
34	тѣсто	dough	*teh ₂ is-to-	B	STR	M
35	мѣхъ	bag	*moysos	B	STR	H
36	вѣкъ	age	*weyk-	B	STR	H
37	сѣверъ	north	*keh ₁ wer-	A	MAX	M
38	невѣста	bride	*new-wed ^h -	C	MOD	M
39	бесѣда	talk	*be-+sed-	A?	MOD	M
40	рѣчь	speech	*rek-	C	STR	H
41	обѣдъ	meal	*ob-+h ₁ ed-	A	MOD	H
42	медвѣдь	bear	*med ^h u+h ₁ ed-	A	MOD	H
43	брѣгъ	shore	*b ^h erg ^h -	C	STR	H
44	ѣсти	to eat	*h ₁ ed-	A	MOD	H
45	грѣти	to heat	*g ^{wh} er-+-eh ₁ -	S	STR	H
46	сѣти	to sow	*seh ₁ y-	A	MOD	H
47	рѣзати	to cut	*wreh ₁ g-	A	MOD	H
48	лѣпити	to stick	*leyp-	B	MOD	H
49	лѣвъ	left	*leh ₂ ywos	B	MOD	H
50	лѣпъ	beautiful	*leyp-	B	MOD	M
51	слѣпъ	blind	*(s)kleyp-?	B	MOD	M
52	лѣха	furrow	*loys-eh ₂	B	STR	H
53	свѣча	candle	*kweyt-	B	STR	H

Continued on next page

#	OCS word	Meaning	PIE root	Grp	LS	Conf
54	сѣть	net	*seh ₁ y-	A	MAX	M
55	стрѣха	roof	*sterh ₃ -	A?	MAX	H
56	бѣсъ	demon	*b ^h oyh ₂ -s-	B	STR	H
57	цѣвь	tube	*koy-w-	B	STR	M
58	плѣва	chaff	*pleh ₁ w-	A	LIM	M
59	грѣхъ	sin	*g ^h rey-?	C	MOD	L
60	дрѣмати	to doze	*drem-+-eh ₁ -	S	STR	M
61	колѣно	knee	*(s)kwel-	C	MOD	M
62	тѣло	body	unknown	?	N/A	L
63	гнѣвъ	wrath	unknown	?	N/A	L
64	желѣзо	iron	substrate?	?	N/A	L
65	пѣсъкъ	sand	unknown	?	N/A	L
66	имѣти	to have	*h ₁ em-+-eh ₁ -	S	MOD	H
67	дрѣво	tree	*dóru/drew-	C	STR	H
68	клѣтъ	cell	*kley-?	B?	STR	L
69	сѣкти	to chop	*sek- (Narten?)	C	LIM	M
70	сѣдло	saddle	*sed-+-lo	A?	MOD	H
71	цѣсарь	emperor	kaisar (Gmc)	loan	STR	H
72	хлѣвъ	cowshed	*hlaiw- (Gmc)	loan	STR	H
73	болѣти	to be sick	*b ^h elH-+-eh ₁ -	S	MOD	M
74	терпѣти	to endure	*terp-+-eh ₁ -	S	STR	M

[†] Borrowing direction debated between Slavic and Germanic.

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